- T_P, the so-called *true positives* are both predicted as true and actually belongs to the `true` class.

Likewise, let N_N be the number of points belonging to the `false` class:

- P_N is the total number of points for which our algorithm predicts the class `false`
- T_N, the so called *true negatives*, is the number of times both the predicted and the actual class of a point are false.

Then we can define accuracy:

$$accuracy = \frac{T_P + T_N}{N_P + N_N} = \frac{T_P + T_N}{N}$$

When the accuracy is 1, we have an algorithm that is always correct.

Unfortunately, except for that ideal case, accuracy is not always a good measure of the quality of our algorithms. Consider an edge case where 99% of the points in a database actually belongs to the true class. Consider the following 3 classifiers:

1. A classifier correctly labeling 100% of the false points and 98,98% of the true ones.
2. A classifier correctly labeling 0,5% of the false points and 99,49% of the true ones.
3. A classifier always returns true as label.